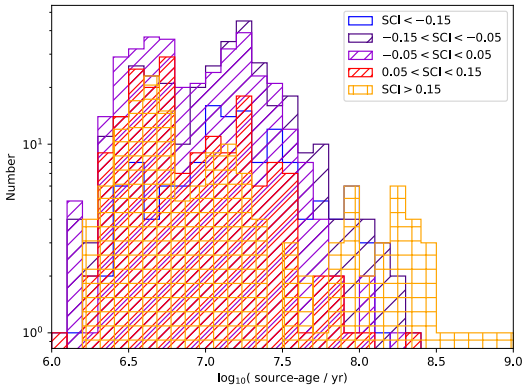
With broadband radio SEDs in place, we are now in prime position for their physical interpretation. For this we make use of RAiSE models, which provide estimates of the source age and jet power across the G4Jy Sample

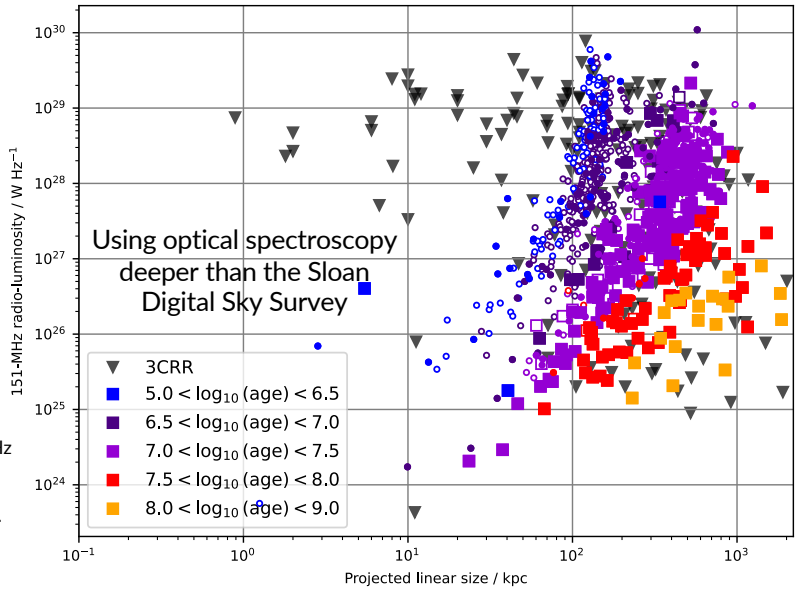
The GLEAM 4-Jy (G4Jy) Sample: V. Broadband spectral analysis

Sarah White (SAAO), Siegfried Gawenda, Giulia Macario, Samuel Hansen, Ross Turner, Thomas Franzen, Joseph Callingham, Randall Wayth, and Kshitij Thorat



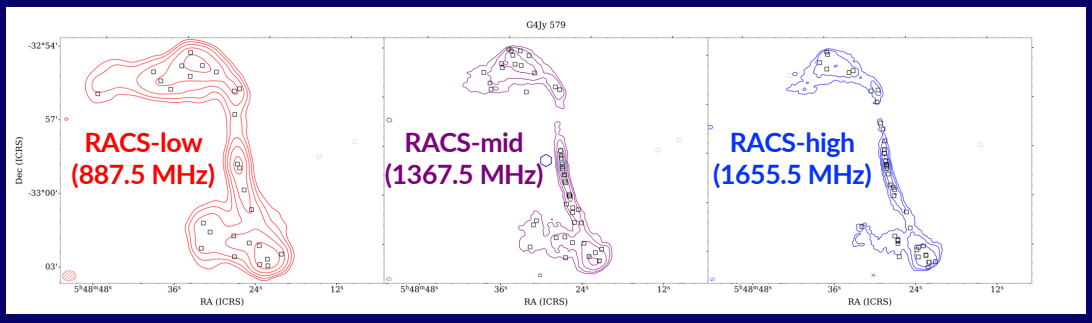
SCI = spectral curvature index
$= \alpha_{low} - \alpha_{mid} = \alpha_{76\ MHz}^{227\ MHz} - \alpha_{151\ MHz}^{1367.5\ MHz}$

As expected, radio galaxies showing the most-positive spectral-curvature are the oldest



Assuming a halo mass of $10^{13.5}$ M_{sol}, G4Jy sources span 0.1 to 891 Myr in age and 10^{36} to 10^{41} W in jet power

We determine RACS (Rapid ASKAP Continuum Survey) and ATCA (Australia Telescope Compact Array) flux-densities via careful visual inspection and manual adjustment



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